ANSWER KEY & MARKING SCHEME SET A

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 90°	Angle sum of a quadrilateral is 360°
2	**(d)** 90°	Angle sum of a quadrilateral is 360°
3	**(b)** 72°	Adjacent angles are supplementary
4	**(b)** 17 cm	Diagonals bisect at right angles; use Pythagoras
5	**(d)** 15 cm	$\sqrt{l^2 + b^2}$
6	**(b)** rhombus	Rhombus property
7	**(c)** meet at right angles	Kite property
8	**(d)** bisect each other	Parallelogram property
9	**(a)** equal non-parallel sides	Property
10	**(d)** two congruent right triangles	Property

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The four interior angles add up to 360° (2).	2
12	$360 - 270 = 90^\circ$ (2).	2
13	They are equal (1); they bisect each other (1).	1+1
14	They bisect each other at right angles (1); they bisect the vertex angles (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let angles be $2x, 3x, 4x, 3x$ (1); $x = 360/12 = 30$ (1); angles 60°, 90°, 120°, 90° (1).	1+1+1
16	Let angles be $1x, 2x, 3x, 4x$ (1); $x = 360/10 = 36$ (1); angles 36°, 72°, 108°, 144° (1).	1+1+1
17	Let angles be $3x, 4x, 5x, 6x$ (1); $x = 360/18 = 20$ (1); angles 60°, 80°, 100°, 120° (1).	1+1+1
	OR: Let angles be $2x, 2x, 3x, 3x$ (1); $x = 360/10 = 36$ (1); angles 72°, 72°, 108°, 108° (1).	1+1+1
18	Let angles be $4x, 5x, 6x, 5x$ (1); $x = 360/20 = 18$ (1); angles 72°, 90°, 108°, 90° (1).	1+1+1
	OR: Let angles be $1x, 3x, 4x, 7x$ (1); $x = 360/15 = 24$ (1); angles 24°, 72°, 96°, 168° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A diagonal splits it into two triangles, each 180° (3). (ii) $360 - 250 = 110^\circ$ (2).	3+2
	OR: (i) Opposite sides and angles equal, diagonals bisect, adjacent angles supplementary (2). (ii) $5x + 10 = 180$, $x = 34$; $P = 88^\circ$, $Q = 92^\circ$, $R = 88^\circ$, $S = 92^\circ$ (3).	2+3
20	(i) Rhombus: equal sides, unequal diagonals meeting at right angles; rectangle: equal diagonals, right angles (3). (ii) Square (2).	3+2
	OR: (i) Kite (1). (ii) $\frac{1}{2} \times 60 \times 40 = 1200$ cm² (2). (iii) Only if both sticks bisect each other (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 4: **Quadrilaterals** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 100°	Angle sum of a quadrilateral is 360°
2	**(c)** 80°	Angle sum of a quadrilateral is 360°
3	**(c)** 65°	Adjacent angles are supplementary
4	**(d)** 10 cm	Diagonals bisect at right angles; use Pythagoras
5	**(a)** square	Regular quadrilateral
6	**(d)** 360°	Two triangles
7	**(a)** rhombus	Square has all sides equal
8	**(d)** 2	Two diagonals
9	**(d)** 360°	One at each vertex
10	**(d)** 90°	Four right angles

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A square has four right angles and opposite sides equal (2).	2
12	Kite: two pairs of equal adjacent sides (1); rhombus: all four sides equal (1).	1+1
13	Opposite 70°, adjacent angles 110° and 110° (2).	2
14	Trapezium (1); e.g. a tabletop shape or a bucket's side (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Half-diagonals 5, 12 form a right triangle (1); side $= 13$ cm (1); perimeter $= 52$ cm (1).	1+1+1
16	Half-diagonals 7, 24 form a right triangle (1); side $= 25$ cm (1); perimeter $= 100$ cm (1).	1+1+1
17	Half-diagonals 8, 15 form a right triangle (1); side $= 17$ cm (1); perimeter $= 68$ cm (1).	1+1+1
	OR: Half-diagonals 9, 12 form a right triangle (1); side $= 15$ cm (1); perimeter $= 60$ cm (1).	1+1+1
18	Adjacent angles are supplementary: $A + B = 180^\circ$, $B + C = 180^\circ$ (2); so $A = C$ (1).	2+1
	OR: $5x = 180$, $x = 36$ (1); $A = 118^\circ$, $B = 62^\circ$ (1); $C = 118^\circ$, $D = 62^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Side 25 cm, perimeter 100 cm, area $\frac{1}{2} \times 30 \times 40 = 600$ cm² (3). (ii) Height $600/25 = 24$ cm (2).	3+2
	OR: (i) Congruent triangles by SAS (3). (ii) 15 m (2).	3+2
20	(i) $\sqrt{150^2 + 80^2} = 170$ cm (2). (ii) 460 cm (1). (iii) 340 cm (2).	2+1+2
	OR: (i) Parallelogram, rectangle, rhombus, square, trapezium, kite with properties (3). (ii) Trapezium (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.